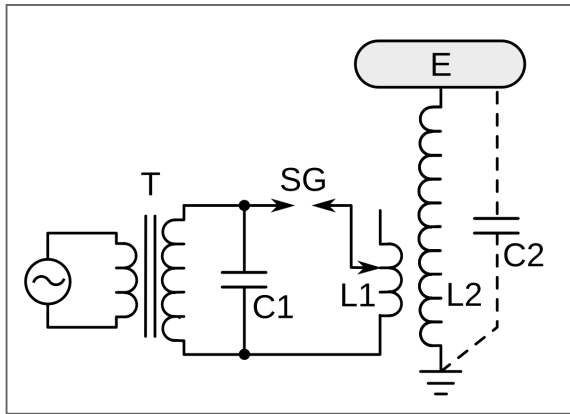


2.7. AC circuits & electromagnetic oscillations



How does a Tesla coil work? ew10

- electric resonant transformer circuit
- Tesla coil with primary and secondary winding, each forming its own LC-circuit
- spark gap (SG) acts as a switch
- need to understand: **LRC-circuits, resonance, and impedance**
- **plan for today:** investigate simply AC-circuits to derive deeper understanding of R, L, and C

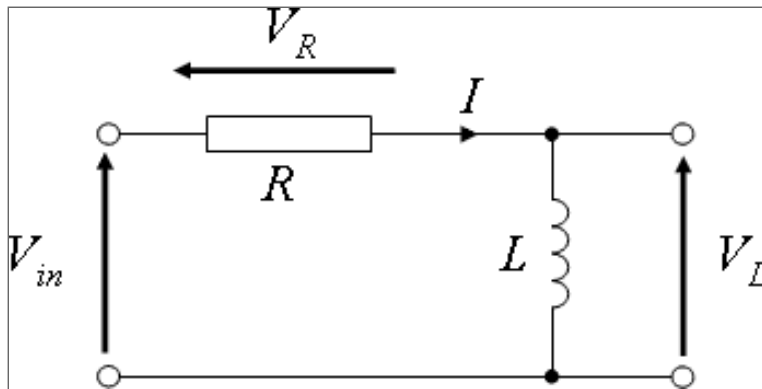


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LR-circuit

em45

- consider a simple DC circuit with the following elements in series:
 - ideal voltage source V_0
 - resistor R
 - inductor L



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LR-circuit: Switching DC supply on

- applying Kirchhoff's loop rule gives

$$V_0 = L \frac{dI}{dt} + IR$$

- rearrange and take integral (0.. I and 0.. t , i.e. current slowly builds up due to self-inductance):

$$\int_0^I \frac{dI}{V_0 - RI} = \int_0^t \frac{dt}{L}$$

- solving the differential equation yields

$$I(t) = \frac{V_0}{R} \left(1 - e^{-tR/L} \right)$$

LR-circuit: Switching DC supply on (cont')

- find V_L :

$$V_L = L \frac{dI}{dt} = \frac{d}{dt} \left(\frac{V_0}{R} \left(1 - e^{-tR/L} \right) \right)$$

$$V_L = L \left(\frac{V_0}{R} \cdot \frac{-R}{L} \cdot e^{-tR/L} \right)$$

$$V_L = -V_0 e^{-tR/L}$$

- **summary:** self-inductance causes:
 - voltage over L to be first opposed to V_0 (Lenz's rule) and decaying over times
 - current building up over time

LR-circuit: Switching DC supply off

- when the DC supply is switched off ($V_0 = 0$) the inductor resists (again) change in current
- Kirchhoff's loop rule becomes

$$L \frac{dI}{dt} + IR = 0$$

- solution for decaying current is:

$$I(t) = I_0 e^{-tR/L}$$

- voltage over inductor is:

$$V_L = L \frac{dI}{dt} = \frac{d}{dt} \left(I_0 e^{-tR/L} \right) = \frac{-LR}{L} I_0 e^{-tR/L}$$

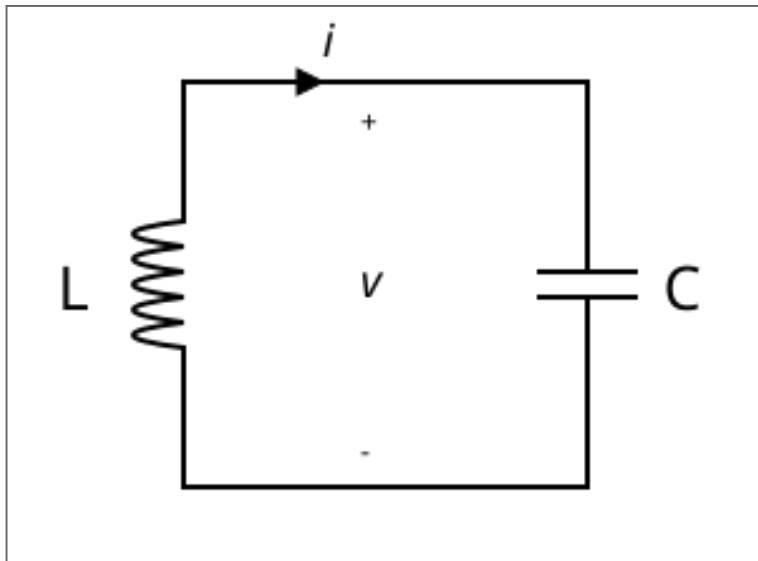
$$V_L = -V_0 e^{-tR/L}$$

- **To summarize** resistor impedes current, i.e. "limits" the current flow; inductor impedes the change in current, i.e. "smooths out" changes in current flow.

LC-circuit: Fun with AC circuit

ew04

What happens if we put L and C in series in a AC circuit?



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LC-circuit: The equations

- applying Kirchhoff's rule leads to:

$$-L \frac{dI}{dt} + \frac{Q}{C} = 0$$

- using $I = -\frac{dQ}{dt}$ (minus because capacitor discharges)

$$\frac{d^2Q}{dt^2} + \frac{1}{LC}Q = 0$$

- second order linear equation describing simple **harmonic oscillator** with general solution:

$$Q(t) = Q_0 \cos(\omega t + \phi) \quad \text{with the angular frequency } \omega = \sqrt{\frac{1}{LC}}$$

- current $I(t)$ is time derivative of charge:

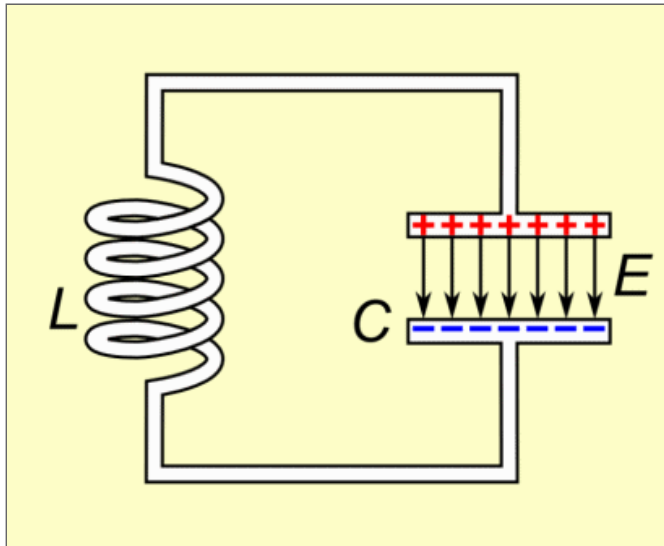
$$I(t) = -\frac{dQ}{dt} = \omega Q_0 \sin(\omega t + \phi) = I_0 \sin(\omega t + \phi)$$

LC-circuit: Electromagnetic oscillations

- the energy oscillates between capacitor (electric field) and inductor (magnetic field):

$$U = U_B + U_E = \frac{Q_0^2}{2C} \sin^2(\omega t + \phi) + \frac{Q_0^2}{2C} \cos^2(\omega t + \phi) = \frac{Q_0^2}{2C}$$

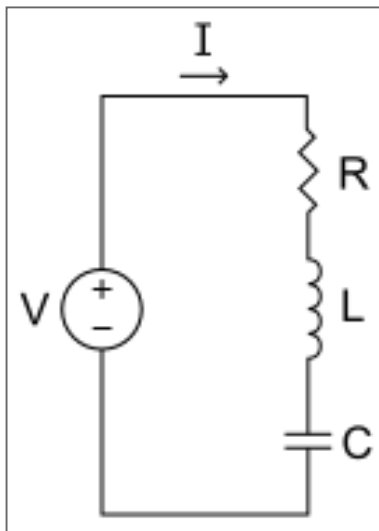
- losses are neglected → **energy conservation**



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LRC-circuit

- adding a resistor
 - introduces damping into oscillations
 - models losses



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LRC-circuit (cont')

sim - damping LRC-circuit

- apply Kirchhoff's loop rule and use $I = \frac{dQ}{dt}$:

$$-L \frac{dI}{dt} - IR + \frac{Q}{C} = 0$$

$$L \frac{d^2Q}{dt^2} + R \frac{dQ}{dt} + \frac{Q}{C} = 0$$

- second-order linear differential equation that describes **damped harmonic oscillator**:
 - $R^2 < \frac{4L}{C}$: Underdamped oscillation with exponential decay
 - $R^2 > \frac{4L}{C}$: Overdamped, i.e. the damping is too strong to allow any oscillations

- $R^2 = \frac{4L}{C}$: Critically damped oscillation with the angular frequency
 $\omega_d = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$

Resistance vs. reactance

ew01 - phase shift

- **LC-circuit:** energy conserved, only exchanged between L and $C \rightarrow$ oscillation $\rightarrow$ phase shift between V and I
- **LRC-circuit:** beyond oscillation, energy dissipated as heat in R
- **new concept:**
 - **resistance** R is independent of frequency and dissipates energy as heat
 - **reactance** (inductive X_L and capacitive X_C) depends on frequency and temporarily stores energy
 - the **phase difference** between voltage and current arises due to reactance
 - reactance & resistance are measured in Ω

Resistor in AC circuit

- follows Ohm's law: $V = IR$
- if current is sinusoidal, i.e. $I = I_0 \cos(\omega t)$, the voltage is:

$$V = (I_0 \cos(\omega t))R = (I_0 R) \cos(\omega t) = V_0 \cos(\omega t)$$

- $\rightarrow$ voltage and current are **in phase**
- the average power is given by

$$\bar{P} = I_{rms}^2 R$$

- resistors can dissipate energy as heat

Inductor in AC circuit

- the voltage across an inductor is: $V + V_{\text{ind}} = V - L \frac{dI}{dt} = 0 \leftrightarrow V = L \frac{dI}{dt}$
- for sinusoidal current $I(t) = I_0 \cos(\omega t)$, the voltage is:

$$V(t) = L \frac{d(I_0 \cos(\omega t))}{dt} = -L\omega I_0 \sin(\omega t)$$

- with $\sin(t) = -\cos(t + \frac{\pi}{2})$

$$V(t) = \omega L I_0 \cos\left(\omega t + \frac{\pi}{2}\right) = X_L I_0 \cos\left(\omega t + \frac{\pi}{2}\right)$$

- the inductive reactance is $X_L = \omega L = 2\pi f L$
- the **voltage leads the current by 90°**
- since X_L **increases with frequency**, inductors resist high-frequency currents more than low-frequency ones

Capacitor in AC circuit

- for a sinusoidal current $I(t) = I_0 \cos(\omega t) = \frac{dQ}{dt}$, charge at capacitor is:

$$Q(t) = \int_0^t dQ = \int_0^t I_0 \cos(\omega t) dt = \frac{I_0}{\omega} \sin(\omega t)$$

- using $\sin \theta = \cos(\theta - \frac{\pi}{2})$, the voltage is:

$$V(t) = \frac{Q}{C} = \frac{1}{\omega C} I_0 \sin(\omega t) = \frac{I_0}{\omega C} \cos\left(\omega t - \frac{\pi}{2}\right) = X_C I_0 \cos\left(\omega t - \frac{\pi}{2}\right)$$

- the capacitive reactance is $X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$
- the **voltage lags the current by 90°**
- since X_C **decreases with frequency**, capacitors resist low-frequency currents more than high-frequency ones

Summary of AC circuit components

- resistor:
 - no phase shift
 - dissipates energy as heat
- inductor:
 - voltage leads current
 - reactance $X_L = \omega L$
- capacitor:
 - voltage lags current
 - reactance $X_C = \frac{1}{\omega C}$
- (ideal) inductors & capacitors store energy, not dissipate it

Impedance

- impedance determines the relationship between voltage and current in AC circuits
- impedance combines resistance and reactance into a complex quantity:

$$Z = R + j(X_L - X_C)$$

- its magnitude is $Z = \sqrt{R^2 + (X_L - X_C)^2}$ and its phase angle is $\tan \phi = \frac{X_L - X_C}{R}$
- impedance in series: $Z_{net} = \sum_i Z_i$
- impedance in parallel: $\frac{1}{Z_{net}} = \sum_i \frac{1}{Z_i}$

Revisiting AC LRC-circuit: Phasor diagrams

sim - Phasor diagram

- in a series LRC-circuit the sum of the voltage drops equals the source voltage: $V = V_R + V_L + V_C$
- in the phasor diagram:
 - V_R is drawn along the positive x-axis (in phase with the current)
 - V_L is drawn 90° ahead of V_R
 - V_C is drawn 90° behind V_R
- the resultant voltage is found by vector addition:

$$V_0 = I_0 Z \quad \text{with} \quad Z = \sqrt{R^2 + (X_L - X_C)^2}$$

Revisiting AC LRC-circuit: Resonance

sim - Resonance

- resonance occurs when the inductive and capacitive reactances cancel:

$$X_L = X_C$$

- the impedance is purely resistive: $Z = \sqrt{R^2 + (X_L - X_C)^2} = R$
- $\rightarrow$ the resonant (angular) frequency is

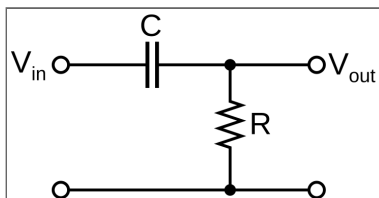
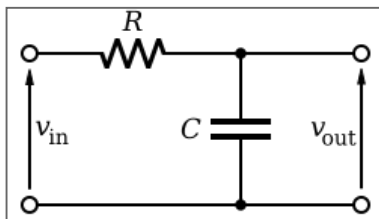
$$\omega_0 = \sqrt{\frac{1}{LC}} \Leftrightarrow f_0 = \frac{1}{2\pi\sqrt{LC}}$$

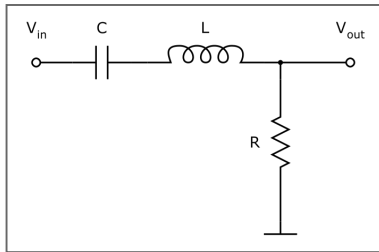
- $\rightarrow$ voltage and current are in phase: $\phi = 0$
- shape of resonance curve depends on the value of R

Filters

ew07 - high/low pass

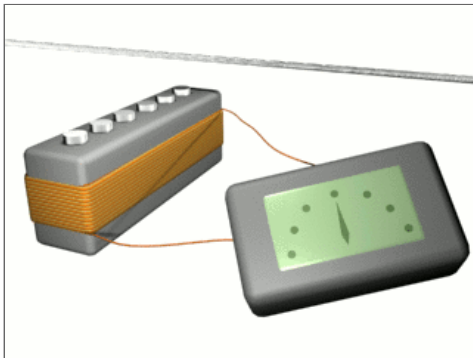
- low-pass filters allow low-frequency signals to pass while attenuating high-frequency ones
- high-pass filters allow high-frequency signals to pass while attenuating low-frequency ones
- band-pass filters allow frequency band to pass while attenuating frequencies below and above the band
- these filters are common in signal processing and audio electronics





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Demo induction, electromagnets, filters, and resonance



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YouTube - bandpass filter in action